HHH-C-575E
September 10, 1987
SUPERSEDING
HHH-C-575D
October 19, 1981

FEDERAL SPECIFICATION

COFFEE, INSTANT

This specification is approved by the U.S. Department of Agriculture (USDA) for use by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers instant coffee; regular and decaffeinated, with or without ascorbic acid, powdered and agglomerated, freeze or spray dried, and commercial or specific blend.

1.2 Classification. Instant coffee shall be of the following types, styles, compositions, and blends, as specified (see 6.1):

Type I - Spray dried, powdered

Style A - Regular Powdered
Composition (1) without ascorbic acid
Composition (2) with ascorbic acid

Style B - Decaffeinated

Type II - Spray dried, agglomerated

Style A - Regular
Style B - Decaffeinated

Type III - Freeze dried

Style A - Regular
Blend (1) - Commercial
Blend (2) - Specific

Style B - Decaffeinated
Blend (1) - Commercial

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 8955

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Federal Specifications

HHH-C-571	-	Coffee
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-G-460	-	Glass Containers, Filled and Closed, Packaging and Packing of

Federal Standard

FED-STD-123	-	Marking for Shipment (Civil Agencies)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, other Federal specifications, standards, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Washington, DC; Philadelphia, PA; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-L-10547	-	Liners, Case and Sheet, Overwrap, Water-vaporproof or Waterproof, Flexible
MIL-L-35078	-	Loads, Unit; Preparation of Semiperishable Subsistence Items: Clothing, Personal Equipment and Equipage; General Specification for

Military Standards

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

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(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulations

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199).

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Defense Personnel Support Center

Procedure for Testing and Certificating of Green Coffee for Use by Contractors in Roasting and Grinding Coffee for the Armed Services of the United States Against Specifications of the Defense Personnel Support Center by the National Coffee Association of the U.S.A.

(Copies may be obtained from Defense Personnel Support Center, Directorate of Procurement and Production, ATTN: DPSC-SPP, 2800 South 20th Street, Philadelphia, PA 19101.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or solicitation for offers shall apply.

American Association of Cereal Chemists

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

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(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

National Research Council

Food Chemicals Codex

(Application for copies should be addressed to the Office of Director, National Academy of Sciences, National Research Council, Washington, DC 20418.)

Association of Vitamin Chemists, Inc.

Methods of Vitamin Assay

(Application for copies should be addressed to the Association of Vitamin Chemists, 3413 Commercial Avenue, Northbrook, IL 60062.)

American Public Health Association

American Public Health Association Fine Sediment Chart

(Copies may be obtained from the Photography Division, Office of Information, U.S. Department of Agriculture, Washington, DC 20250.)

New York Coffee, Sugar, and Cocoa Exchange

New York Coffee, Sugar, and Cocoa Exchange Standard

(Application for copies should be addressed to the New York Coffee, Sugar, and Cocoa Exchange, Four World Trade Center, 8th Floor, New York, NY 10048.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1120, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Bid sample approval. Unless otherwise specified (see 6.1), six individually packaged samples of Type I, Type II, or Type III, Style A product containing at least 2 ounces each, representative of the product which the bidder proposes to furnish (except that, the bid sample of Type I, Style A, Composition (2), need not contain ascorbic acid unless otherwise specified), shall be submitted to the contracting officer prior to bid opening. Bid samples shall be evaluated for palatability acceptance by a panel of USDA coffee specialists. Approved bid samples shall be used as standards in determining the acceptance of delivery product for palatability as specified in 3.4.1. The approval of any bid sample for palatability will not constitute approval of the bid sample in meeting the other requirements of this specification.

3.2 Materials. All materials shall be edible, clean, sound, and wholesome. For military, the raw or green coffee bean utilized for both Types I, II, and III coffees shall be processed, packaged, and packed within the United States subsequent to date of award. The roasted and ground coffee from which the instant coffees are derived shall be made from coffee received in the raw or green bean state. Raw or green bean coffee is defined as the dried seed free from external layers (skin, pulp, mucilage, parchment, and silver skin) before roasting.

3.2.1 Coffee, (Types I and II). The green coffee shall be of such growth or growths and grade as to produce an end product cup quality free of Rio or other objectionable flavors. The coffee extracts shall be dried with the least practical delay. The dried product shall be powder, and may be produced from dried agglomerated particles.

3.2.2 Coffee, (Type III, Style A, Blend (1) and Type III, Style B, Blend (1). The green coffee for commercially freeze dried product shall be of such growth or growths and grade as to produce an end product cup quality free of Rio or other objectionable flavors.

3.2.3 Coffee, (Type, III, Style A, Blend (2)). Coffee, specific blend, freeze dried, shall be a blend of green coffees consisting only of Brazil, Colombian Excelso, and Central American Milds in a proportion of not more than 70 percent Brazil or not more than 70 percent Colombian and Central American Milds, of which not less than 50 percent shall be Colombian Excelso. Brazils shall be Type I-A, Colombian Excelso shall be Type I-B, respectively, as specified in HHH-C-571. Central American Milds shall be current crop and of the following growths and description: 1. Prime Washed Guatemalan, 2. Prime Washed Mexican, 3. Central Standard Washed Salvadoran.

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Grade of these Central American Milds shall be not lower than maximum number of imperfections below type permitted for these coffees by the New York Coffee, Sugar, and Cocoa Exchange. Coffee shall be sweet in the cup; good roasting quality; and a bean size in accordance with standards established by New York Coffee, Sugar, and Cocoa Exchange. The green coffees shall be tested and certified in accordance with Procedure for Testing and Certificating of Green Coffee for use by Contractors in Roasting and Grinding Coffee for the Armed Services of the United States Against Specifications of the Defense Personnel Support Center by the National Coffee Association of the U.S.A.

3.2.4 Coffee, decaffeinated (Types I, II, and III, Style B). Decaffeinated coffee, shall be decaffeinated using Good Manufacturing Practice, and following Food, Drug, and Cosmetic Act guidelines.

3.2.5 Ascorbic acid. Ascorbic acid used in the Type I, Style A, Composition (2) product shall be in accordance with the requirements of Food Chemicals Codex, and shall be added to coffee after drying.

3.3 Processing.

3.3.1 Extracting and drying equipment. All utensils and extracting and drying equipment used shall be clean and free from objectionable odors. The coffee extracts shall be dried with the least practicable delay.

3.3.2 Freeze drying processing. The Type III, Style A and B product shall be a soluble coffee which is processed mainly by sublimation during drying.

3.4 Finished product. The finished product requirements shall be as specified in Table I and 3.4.1.

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TABLE I. Finished product requirements

	Percent moisture	Percent caffeine	Sediment	Particle size 1/	Added ascorbic acid
Type I, Style A, Composition (1) without ascorbic acid	3.0 maximum	3.2 minimum	Not more than 2.0 milligrams on disk in the American Public Health Assoc. Fine Sediment Chart	--	--
Type I, Style A, Composition (2) with ascorbic acid	Same as above	Same as above	Same as above	--	Not less than 15 milligrams per 2.2 grams
Type I or II, Style B	5.0 maximum	0.3 maximum	Same as above	--	--
Type II, Style A	5.0 maximum	3.2 minimum	Same as above	--	--
Type III, Style A, Blend (1) or (2)	2.6 maximum	2.2 minimum	Same as above	U.S. Standard Sieve No. / Percent by weight remaining: 8 / None; 12 / 20 max.; 40 / 80 ± 15; Pan / 15 max.	--
Type III, Style B, Blend (1)	5.0 maximum	0.3 maximum	Same as above	--	--

1/ Certificate of conformance shall be supplied at the request of contracting officer for Type III, Style A.

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3.4.1 Palatability. Delivery samples of the finished product shall be cup tested by a panel of experienced USDA coffee specialists in accordance with 4.4.5. Coffee shall possess the flavor and aroma of the blend procured. Coffee which passes the sensory tests shall conform to all other requirements as specified herein.

3.5 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110: "Current Good Manufacturing Practice in Manufacture, Processing, Packaging or Holding of Human Foods"; and the plant sanitation requirements of the appropriate Government inspection agency.

3.6 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this document.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.3.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

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4.3.1.2 Envelope material examination. Conformance of laminated envelope material to requirements of 5.1.1, shall be ascertained by examination of invoices or other valid documentation from the supplier of the envelope material. Nonconformance shall indicate unacceptable material and use of such shall be cause for rejection of the involved end item.

4.3.2 In-process examination. Examination shall be performed to determine conformance to 3.2.3, as concerns blending of Type III, Style A, Blend (2), green coffees and 3.3, as concerns methods of processing. Nonconformance shall be cause for rejection of the involved finished product.

4.3.3 End item inspection.

4.3.3.1 End item examination. The finished product shall be examined for the presence of foreign material (e.g., glass, wood, paint, filth, insects, metal, etc.). The sample unit shall be one filled and sealed envelope, or one flexible unit pack, or one glass container, as applicable. The lot size shall be expressed in units of envelopes, or flexible unit packs, or glass containers, as applicable. The inspection level shall be S-2. The presence of any foreign material shall be cause for rejection of the lot.

4.3.3.2 End item testing. The finished product shall be tested for the characteristics listed in Table II. The sample for testing shall be a 4-ounce composite of the finished product derived from the number of primary containers indicated by inspection level S-2 except that for envelopes, the sample shall be composited from the contents of a sufficient number of envelopes to yield 4 ounces. When product with ascorbic acid is to be tested, a 2-ounce sample of unfortified product shall accompany the delivery product samples. The lot size shall be expressed in units of primary containers. Any test failure shall be cause for rejection of the lot. In addition, the finished product shall be tested for particle size (Type III only) and palatability, as specified in 4.3.3.3 and 4.3.3.4.

TABLE II. End item tests

Characteristic	Requirement	Test method	No. of determinations	Results reported as
Moisture content	Table I	4.4.1	1	Nearest 0.1 percent
Caffeine content	Table I	4.4.1	1	Nearest 0.1 percent
Sediment	Table I	4.4.2	1	Pass or fail
Ascorbic acid	Table I	4.4.4	1	Nearest 1 mg/2.2 grams of product

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4.3.3.3 Particle size testing (Type III only). The finished product shall conform to the requirement specified in Table I when tested as specified in 4.4.3. The sample unit shall be one filled and sealed primary container. The lot size shall be expressed in units of primary containers. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 1.5 test failures per hundred units.

4.3.3.4 Palatability testing. The finished product shall conform to the requirements specified in 3.4.1 when tested as specified in 4.4.5. The sample unit shall be one filled and sealed primary container. The lot size shall be expressed in units of primary containers. The inspection level shall be S-2 and the AQL shall be 1.0 test failures per hundred units.

4.3.4 Packaging inspection.

4.3.4.1 Primary and intermediate container examination. Examination of primary and intermediate containers shall be in accordance with the United States Standards for Condition of Food Containers.

4.3.4.2 Net weight examination. Primary containers (filled envelopes, flexible unit packs, or glass containers, as applicable) shall be examined for the net weight defects listed in Table III. The lot size shall be expressed in units of primary containers. The sample unit shall be one filled primary container. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

TABLE III. Examination for net weight 1/

Category	Defect
Minor	
201	Less than 1.4 grams, 1.5 grams, 1.7 grams, or 2.1 grams (for the respective 1.5-gram, 1.6-gram, 1.8-gram or 2.2-gram unit pack)
202	Less than 3.8 ounces or 7.6 ounces (for the respective 4-ounce and 8-ounce unit pack)

1/ Lot shall be rejected if sample data indicate a lot average net weight less than specified net weight.

4.3.4.3 Packing examination. Packing defects shall be classified in accordance with Table IV. The lot size shall be expressed in units of shipping containers. The sample unit shall be one filled shipping container. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects.

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TABLE IV. Examination of packing

Category		Defect
Major	Minor	
101		National stock number, item description, contract number, or date of pack markings on shipping container missing, incorrect, or illegible.
	201	Other required markings on shipping container missing, incorrect, or illegible.
102		Any component missing, damaged, or not as specified.
103		When applicable, waterproof case liner missing or not protected from metal fasteners.
	202	Number of unit packs per intermediate box or shipping container less than specified.
	203	Weight of product per shipping container greater than specified.
	204	When applicable, reinforcement with nonmetallic strapping or tape has not been used.
	205	Not snug pack.

4.3.4.4 Examination of glass containers. Examination of glass containers shall be in accordance with the quality assurance provisions of PPP-G-460.

4.3.4.5 Unit load examination. Unit loads shall be examined in accordance with the quality assurance provisions in MIL-L-35078.

4.3.4.6 Envelope leakage testing (Type I and Type II, 2.2-gram content only). Envelopes shall be tested as specified in 4.4.6 for conformance to the leakage requirement in 5.1.1.1.1. The lot size shall be expressed in units of envelopes. The sample unit shall be one filled and sealed envelope. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.0.

4.4 Methods of inspection.

4.4.1 Chemical analyses. Chemical analyses for moisture and caffeine content shall be conducted in accordance with the following methods from Official Methods of Analysis of the Association of Official Analytical Chemists with the exception specified in footnote 1/ below.

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Test	Source	Method
Moisture 1/	Chapter: Coffee and Tea	Moisture: Loss on Drying (vacuum oven)
Caffeine (Type I, Type II, and Type III, except Style B)	Chapter: Coffee and Tea	Caffeine Bailey-Andrew Method
Caffeine (Style B)	Chapter: Coffee and Tea	Caffeine-Chromatographic-Spectrophotometric Method

1/ Temperature during drying shall be 140° to 158°F (60° to 70°C) under a pressure of 3.9 inch to 5.9 inch Hg.

4.4.2 Sediment test. Weigh 5.0 grams of instant coffee into a 250 mL beaker. Add 200 mL of hot distilled water (temperature not below 185°F). Stir until coffee is completely dissolved. Filter hot solution through sediment filtering material held in a sediment filtering apparatus. The filtering apparatus shall be so constructed that an effective filtering area 1-1/8 inch in diameter of filtering material is exposed, and that no sample being filtered shall bypass the filter material. The filter area shall be unobstructed except for the wire mesh needed to support the filter. The filter material used shall be Johnson & Johnson cotton lintine sediment testing filter material or material giving equivalent results under conditions of the test. The temperature of the coffee solution shall not be less than 140°F at time of filtration. Wash the beaker and sediment tester with small portions of hot water so as to transfer all sediment to the surface of the filter material, and to wash out all of the soluble coffee from the filter. Total wash water shall not exceed 200 mL. Remove filter containing sediment from apparatus, dry, and compare with photographs of standard sediment disks in the American Public Health Association Fine Sediment Chart.

4.4.3 Particle size (type III product) test. Assemble sieves and collecting pan in the following order from top to bottom; U.S. Standard Sieve No. 8, No. 12, No. 40, and pan. Place the weighed contents (to nearest gram) of a sample package on the top sieve and attach lid. Place assembly in a ro-tap or equivalent mechanical shaking device and shake for 1 minute, in a room in which the relative humidity does not exceed 40 percent. Weigh product remaining on each sieve and in the collecting pan respectively. Calculate results in percentage and report results to nearest 1 percent.

4.4.4 Ascorbic acid. The amount of ascorbic acid in the fortified instant coffee shall be calculated by the difference from the assay values of the unfortified and the fortified product, divided by the percent of recovery. The method used shall be essentially that of Loeffler and Ponting described in Methods of Vitamin Assay of the Association of Vitamin Chemists, Inc. An

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aqueous-acid extract shall be made of the material being tested, the extract filtered and an aliquot of the filtrate taken for reaction with the indophenol reagent. The latter is present in excess so that the residual color can be measured in a photometric colorimeter. The method of recovery shall be checked by running an assay on the unfortified samples with a known amount of added ascorbic standard. The increment shall be compared with a known amount of added ascorbic standard. The increment shall be compared with the results obtained on the fortified sample and shall be corrected for percent recovery in order to account for the loss incurred in the procedure.

4.4.4.1 Reagents.

Two percent by weight metaphosphoric acid (HPO_3) (C.P.) dissolved in redistilled H_2O.

This solution is stored in the refrigerator when not in use and is prepared fresh about every 3 or 4 days. The dye solution is made by dissolving 13 to 17 mg of 2, 6 dichlorophenolindophenol in redistilled H_2O and diluting to one liter. This solution is standardized by setting the instrument to 100 percent transmittance at 520 nm with 10 mL of redistilled water and determining the 15 second reading given by a tube containing a 1 mL of 2 percent HPO_3 and 9 mL of dye solution. This reading is G_1 and should be about 30.

4.4.4.2 Methods.

(a) A representative sample is dissolved in a volume of 2-percent HPO_3 and filtered through No. 12 Whatman filter paper.

	Sample weight	Volume 2-percent HPO_3
Unfortified	5 grams	1 liter
Fortified	5 grams	1 liter

The sample size may have to be adjusted, depending upon the concentration of ascorbic acid and the interference from color with larger samples.

(b) One mL aliquots of the filtrate are measured into each of three matched colorimeter tubes. To one is added 9 mL of redistilled water, the contents of the tube mixed by inversion, the tube inserted in the instrument and the instrument adjusted to 100 percent transmittance (520 nm). This corrects for any color or turbidity which may be present in the sample filtrate.

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(c) To each of the other tubes (one at a time) is added 9 mL of the previously standardized dye solution. The 9 mL of dye solution is added from a rapid-delivery pipette, emptying in less than 5 seconds. The contents of the tube are mixed by inversion and a reading taken 15 seconds after the start of the dye addition. This reading is G_2.

4.4.4.3 Calculations. K (L_1 - L_2) x Vol. of HPO_3 = mg ascorbic acid per gram of sample. Divide the result by 2.2 grams to obtain ascorbic acid results per 2.2 grams. Report results to the nearest milligram per 2.2 grams. L_1 = log 100 - log G_1, or color density corresponding to galvanometer reading of dye. L_2 = log 100 - log G_2, or color density corresponding to galvanometer reading of sample. The instrument constant formula relates color to milligrams of ascorbic acid:

$$K = \frac{\text{mg ascorbic acid per mL}}{L_1 - L_2}$$

This is obtained by determining the galvanometer reading for solutions of ascorbic acid varying in concentration from 5 to 50 gamma per mL. Only that range of galvanometer readings which gives a constant K value may be used in assay work unless a calibration curve is established.

4.4.5 Palatability test. The sample shall be prepared for palatability testing by adding freshly boiled water at a temperature of 185°F following label directions.

The palatability determination shall be made when the beverage is at a temperature of 145°F. A Type I, Type II, or Type III, Style A sample not having the same flavor characteristics as the bid sample or a sample not characteristic of the specified blend shall be considered a test failure.

4.4.6 Leakage test. The filled and sealed envelopes shall be tested for leakage by placing them in a dry desiccator, or similar apparatus, and subjecting them to a vacuum of not less than 26 inches of mercury for 30 seconds. Any envelope that does not swell to form a tightly distended package having at least one distorted edge during the test shall be recorded as a leaker. After vacuum testing, the envelopes shall be visually inspected for evidence of delamination and for seal separation. Any leakage, any delamination, or any seal separation of more than 1/16 inch from the product edge of any seal shall be recorded as a test failure.

5. PREPARATION FOR DELIVERY

5.1 Preservation. The product shall be preserved in accordance with Level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit packing.

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5.1.1.1.1 Two and two tenths gram quantity. A net weight of 2.2 grams of Type I or Type II product shall be unit packed in an envelope. A minus tolerance of 0.1 gram will be allowed in the net weight of any unit pack, provided the average net weight of the unit packs in any lot is not less than the specified net weight when examined in accordance with Table III. The envelope shall be made from a heat-sealable, laminated material, one lamina of which shall be a minimum of 0.00035 inch thick aluminum foil. The envelope shall be dull yellow or light brown in color. The envelope shall be heat sealed along all four sides with the seals not less than 3/16 inch wide. The sealed envelope shall not leak when tested in accordance with 4.3.4.5. The filled and sealed envelope shall have dimensions of not more than 3-1/2 inches in length, 2-1/2 inches in width, and 1/8 inch in thickness.

5.1.1.1.2 Four-ounce and 8-ounce quantity. A net weight of 4 ounces or 8 ounces of product shall be unit packed in a heat-sealable, laminated material, one lamina of which shall be aluminum foil, commercially used for the product. A minus tolerance of 0.2 ounce for 4-ounce and 0.4 ounce for 8-ounce packages will be allowed in the net weight of any unit pack, provided the average net weight of the unit packs in any lot is not less than the specified net weight when examined in accordance with Table III. Each flexible unit pack shall be securely closed by heat sealing.

5.1.1.2 Intermediate packing.

5.1.1.2.1 Four-ounce and 8-ounce quantity. Forty-eight 4-ounce or twenty-four 8-ounce unit packs of product of one description only, unit packed as specified in 5.1.1.1.2, shall be packed in a snug-fitting fiberboard box. Each box shall be securely closed.

5.1.1.2.2 Two and two tenths-gram quantity. Two hundred or one thousand (see 6.1) 2.2-gram unit packs of product of one description only, unit packed as specified in 5.1.1.1.1, shall be packed in a snug-fitting paperboard or fiberboard box. Each box shall be securely closed. When specified (see 6.1), two thousand 2.2-gram unit packs of one product of one description only, shall be packed directly into the Level C shipping container specified in 5.2.3.1 without first being packed into intermediate boxes.

5.1.2 Level C. The product, in net weight quantities of 1.5 grams, 1.6 grams, 1.8 grams, 4 ounces, or 8 ounces shall be furnished in commercial packaging. The package shall afford adequate protection against deterioration and physical damage during shipment from the supply source to the first consuming activity. A minus tolerance of 0.1 gram for 1.5-gram, 1.6-gram, or 1.8-gram unit packs or minus tolerance of 0.2 ounce or 0.4 ounce for 4-ounce or 8-ounce unit packs respectively will be allowed in the net weight of any individual commercial package, provided the average net weight of the commercial packages in any lot is not less than the applicable specified net weight when examined in accordance with Table III.

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5.2 Packing. The product shall be packed Level A, B, or C, as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Two and two tenths-gram quantities. Not more than 40 pounds of product of one description only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed, closed, and reinforced in accordance with Style RSC, Grade V2s of PPP-B-636. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament-reinforced tape in accordance with the Appendix of PPP-B-636.

5.2.1.2 Four-ounce and 8-ounce quantities. Ninety-six 4-ounce or forty-eight 8 ounce unit packs of product of one description only, preserved as specified in 5.1, shall be packed in accordance with 5.2.1.1. The contents of each box shall be enclosed in a waterproof case liner, fabricated and closed in accordance with MIL-L-10547 for subsistence items. When the box is closed or the manufacturer's joint is made with stitches or staples, the exposed metal fastener on the inside of the box shall be covered with pressure-sensitive tape or other means to provide physical protection to the case liner.

5.2.1.3 Glass containers. The product, packaged commercially in accordance with 5.1.2, shall be packed in accordance with the applicable Level A requirements specified in PPP-G-460.

5.2.1.4 One and one-half-gram, 1.6-gram, and 1.8-gram quantities. Not more than 40 pounds of product of one description only, packaged commercially in accordance with 5.1.2, shall be packed in accordance with 5.2.1.1

5.2.2 Level B.

5.2.2.1 Two and two tenths-gram quantities. Not more than 40 pounds of product of one description only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed, closed, and reinforced in accordance with Style RSC, Grade V3c, V3s, or V4s of PPP-B-636. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament-reinforced tape in accordance with the Appendix of PPP-B-636.

5.2.2.2 Four-ounce and 8-ounce quantities. Ninety-six 4-ounce or forty-eight 8-ounce unit packs of product of one description only, preserved as specified in 5.1, shall be packed as specified in 5.2.2.1. The box shall be fitted with a waterproof case liner as specified in 5.2.1.2.

5.2.2.3 Glass containers. The product, packaged commercially in accordance with 5.1.2, shall be packed in accordance with the appliable Level B requirements specified in PPP-G-460.

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5.2.2.4 One and one-half-gram, 1.6-gram, and 1.8-gram quantities. Not more than 40 pounds of product of one description only, packaged commercially in accordance with 5.1.2, shall be packed as specified in 5.2.2.1.

5.2.3 Level C.

5.2.3.1 Two and two tenths-gram quantities. Product of one description only, preserved as specified in 5.1, shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall comply with the Uniform Freight Classification or National Motor Freight Classification as applicable, except fiberboard containers shall be closed in accordance with Method II as specified in the Appendix of PPP-B-636.

5.2.3.2 Four-ounce and 8-ounce quantities. Ninety-six 4-ounce or forty-eight 8-ounce unit packs of product of one description only, preserved as specified in 5.1, shall be packed as specified in 5.2.3.1. When specified (see 6.1), twenty-four 4-ounce unit packs shall be packed directly into the shipping container without the intermediate packing specified in 5.1.1.2.

5.2.3.3 Glass containers. The product, packaged commercially in accordance with 5.1.2, shall be packed in accordance with the applicable Level C requirements specified in PPP-G-460.

5.2.3.4 One and one-half-gram, 1.6-gram, and 1.8-gram quantities. Not more than 40 pounds of product of one description only, packaged commercially in accordance with 5.1.2, shall be packed as specified in 5.2.3.1. When specified (see 6.1), one thousand 1.5-gram, 1.6-gram, or 1.8-gram commercial unit packs shall be the packing quantity.

5.3 Unit loading. When specified (see 6.1), instant coffee, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Labeling and marking.

5.4.1 Two and two tenths-gram envelopes. Printed matter shall be in contrasting color (see 5.1.1.1.1), to allow easy and legible reading. The following information shall be included:

NAME OF PRODUCT (in letters larger
than any others used for the label)
2.2 grams net
With ascorbic acid (when applicable)

Dissolve contents in
1/3 canteen cup (8 fluid ounces) of hot water. Stir.
Name of contractor
Address of contractor

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5.4.2 One and one-half-gram, 1.6-gram, 1.8-gram, 4-ounce, and 8-ounce unit packs. Labeling shall be in accordance with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder and shall include directions for use.

5.4.3 Intermediate pack. Each intermediate pack shall be marked with the following information: Item description, number of unit packs per intermediate pack, and quantity (net weight) of product per unit pack.

5.4.4 Shipping containers. Shipping containers shall be marked in accordance with FED-STD-123 or MIL-STD-129, as applicable.

5.4.5 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Purchasers shall select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, style, composition, blend, and unit pack quantity (net weight) of product required (see 1.2 and section 5).
(c) When other than six bid samples are desired (see 3.1).
(d) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(e) Whether two hundred or one thousand 2.2-gram unit packs are to be packed in an intermediate box (see 5.1.1.2.2).
(f) When two thousand 2.2-gram unit packs are required for shipment to ration assembler (see 5.1.1.2.2).
(g) When twenty-four 4-ounce flexible unit packs per Level C shipping container is required (see 5.2.3.2).
(h) When one thousand 1.5-gram, 1.6-gram, or 1.8-gram unit packs per Level C shipping container is required (see 5.2.3.3).
(i) Type and class of unit load when unit loading is specified (see 5.3).

6.2 Appropriate level of pack (military agencies). Based on conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15, NAVSUPINST 4030.28, AFR 71-6, MCO 4030.33A/DLAR 4145.7, as applicable.

6.3 Classification. HHH-C-575D designations and corresponding new designations in HHH-C-575E shown below.

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HHH-C-575D

Type I, Style 1, Composition (1)
Type I, Style 1, Composition (2)
Type I, Style 2
Type I, Style 3
Type I, Style 2
Type II, Class A
Type II, Class B
None

MILITARY INTERESTS:

Custodians

Army - GL
Navy - SA
Air Force - 50

Review Activities

Army - MD, TS
Navy - MC
DP - SS

HHH-C-575E

Type I, Style A, Composition (1)
Type I, Style A, Composition (2)
Type I, Style B
Type II, Style A
Type II, Style B
Type III, Style A, Blend (1)
Type III, Style A, Blend (2)
Type III, Style B, Blend (1)

CIVIL AGENCY COORDINATING ACTIVITIES:

USDA - MRD
HHS - NIH
VA - OSS

PREPARING ACTIVITY:

Army - GL

Project No. 8955-0056

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